

Following above, in the admin panel, there is an Action box. In the Action box, there is an option “Delete selected ...”. Please tell me where the phrase “Delete selected ...” is?

The phrase “**Delete selected ...**” in the Django admin panel comes from Django’s built-in admin actions. Specifically, it’s defined in the Django source code as the default action for model deletion.



Where It Comes From

- The phrase is generated by Django’s internal translation system.
- It’s defined in the `django/contrib/admin/actions.py` file.
- The actual function is called `delete_selected`, and its label is set using:

```
delete_selected.short_description = _("Delete selected %(verbose_name_plural)s")
```

- The `%(verbose_name_plural)s` part is dynamically replaced with the plural name of the model (e.g., “financials”, “operations”, “assets”).



How It Appears in the Admin Panel

- When you register a model with `ModelAdmin`, Django automatically includes this action unless you override it.
- It shows up in the **Action dropdown** on the change list page.
- If you select rows and choose “Delete selected financials”, Django will prompt for confirmation before deletion.



Can You Customize or Remove It?

Yes! You can:



Remove it:

```
class FinancialAdmin(admin.ModelAdmin):  
    actions = [] # disables all actions including delete
```



Customize it:

You can override the `delete_selected` function or write your own with a different label.

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Would you like to rename “Delete selected...” to something more user-friendly like “Remove selected entries”? I can show you how to do that next.

Following above, where is “delete_selected.short_description = _("Delete selected %(verbose_name_plural)s)”? Is it located under assignment/config/? Or I have no access right to it? Please advise.

Great question, SIU — and you're absolutely right to wonder where that phrase is defined.



Where “Delete selected ...” Comes From

The line:

```
delete_selected.short_description = _("Delete selected %(verbose_name_plural)s")
```

is **not** located in your project folder like `assignment/config/`. Instead, it's part of **Django's internal source code**, specifically in:

```
django/contrib/admin/actions.py
```

This file defines the built-in `delete_selected` admin action.



Do You Have Access to It?



Yes, but only if:

- You installed Django via pip (e.g., `pip install django`)
- You browse into your Python environment's `site-packages` folder



No, if:

- You're only working inside your Django project folder (`assignment/`)
- You're using Django as a managed dependency and don't explore the installed packages



Can You Customize It?

Yes! You can override or remove the default action in your own admin class:



Remove it:

```
class FinancialAdmin(admin.ModelAdmin):  
    actions = [] # disables all actions including delete
```

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Customize it:

You can define your own delete action with a custom label and logic.



Summary

- The phrase is defined in Django's core code, not your project.
- You can't see or edit it directly unless you browse Django's source files.
- You **can override or replace** it in your own admin classes.

Would you like help writing a custom delete action with a confirmation message or a different label like "Remove selected entries"? I can show you how.

在電腦學中，格式化資料是什麼意思？

當然！這是一個在電腦科學中非常基礎且重要的概念。簡單來說：

****格式化資料 (Formatted Data) **** 是指按照****預先定義好的結構、規則或標準****來組織和排列的資料。

它的核心目的是讓資料能夠被****高效、準確地交換和處理****，無論是由不同的電腦程式、系統，還是不同的人來讀取。

一個簡單的比喻

想像一下你寫一封信：

* ****未格式化的資料****：就像你把所有內容（收件人、地址、正文、問候語）全部寫在一行，沒有分段和標點。雖然人類可能勉強能看懂，但電腦會很難分辨哪部分是地址，哪部分是正文。

* ****格式化的資料****：就像標準的信件格式，收件人、地址、日期、正文、署名都出現在約定俗成的位置。無論是誰或哪個系統，都能清楚地理解每個部分的意義。

格式化的主要特點與好處：

1. ****結構化****：資料被分成不同的欄位、記錄或層次。

* ****例如****：一個「員工」資料，會被結構化成「姓名」、「員工 ID」、「部門」、「薪水」等欄位。

2. ****標準化****：遵循一個公開的、通用的標準，確保互通性。

* ****例如****：JSON、XML、CSV 就是通用的資料格式標準。

3. ****可被機器解析****：這是格式化最重要的目的。程式可以根據已知的格式規則，自動地、準確地從中提取所需的資訊。

4. ****可驗證****：可以根據格式規則來檢查資料是否完整、有效。

* ****例如****：可以檢查一個標記為「電子郵件」的欄位是否符合電子郵件的格式。

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常見的格式化資料範例：

1. **JSON**

目前網路和應用程式間最流行的資料交換格式。它使用「鍵-值」對和陣列來組織資料，對人類和機器都非常友好。

```
```json
{
 "employees": [
 {
 "name": "張小明",
 "employeeId": 001,
 "department": "工程部",
 "isManager": true
 },
 {
 "name": "王美美",
 "employeeId": 002,
 "department": "行銷部",
 "isManager": false
 }
]
}
```
```

2. **XML**

早期非常流行的標記語言，使用標籤來定義資料結構。

```
```xml
<employees>
 <employee>
 <name>張小明</name>
 <employeeId>001</employeeId>
 <department>工程部</department>
 <isManager>true</isManager>
 </employee>
 <employee>
 <name>王美美</name>
 <employeeId>002</employeeId>
 <department>行銷部</department>
 <isManager>false</isManager>
 </employee>
</employees>
```
```

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3. **CSV**

用逗號分隔的純文字格式，非常適合用來表示表格資料，常見於試算表和資料庫匯出。

```
```csv
name,employeeId,department,isManager
張小明,001,工程部,true
王美美,002,行銷部,false
```
```

4. **YAML**

一種易於閱讀的資料序列化格式，常用於設定檔。

```
```yaml
employees:
- name: 張小明
 employeeId: 1
 department: 工程部
 isManager: true
- name: 王美美
 employeeId: 2
 department: 行銷部
 isManager: false
```
```

5. **資料庫表格**

關聯式資料庫本身就是高度格式化的資料，有明確的欄位名稱、資料型別（整數、字串、日期等）和表之間的關係。

與「未格式化資料」的對比

* **未格式化資料**：

- * **純文字文件**：一篇小說、一篇文章。電腦知道它是文字，但不理解其內在結構。
- * **圖片**：一張 .jpg 或 .png 圖片，電腦知道它是影像，但不直接知道圖片裡有什麼物件。
- * **影音檔案**：一段 .mp3 或 .mp4 檔案。
- * **二進位亂碼**：沒有任何結構可言的資料流。

總結

在電腦科學中，**格式化資料就是為資料穿上一件標準的「制服」**，讓程式能夠像機器人一樣，按照統一的規則快速、無誤地處理它們。它是實現系統整合、資料交換和自動化的基石。

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